Behavioral Economics of AI: Experimental Evidence on Preference Signatures of Large Language Models

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Abstract

We apply established behavioral economics experimental methods to elicit preference parameters from large language models (LLMs). Our key methodological innovation is a **Three-Arm Treatment Design** that separately identifies three distinct response systems: (i) the model's operational policy (Self arm), (ii) its learned beliefs about human behavior (Human arm), and (iii) its knowledge of normative economic theory (Normative arm). This design solves a fundamental identification challenge in LLM behavioral research by disentangling preference-like behavior from simulation and knowledge retrieval. Testing seven frontier models from four providers, we find: (1) substantial cross-model heterogeneity in risk aversion, loss aversion, and time preferences, with effect sizes exceeding typical human variation; (2) provider-level "behavioral fingerprints" consistent with RLHF training objectives—OpenAI models are risk-seeking while Anthropic models exhibit no present bias; (3) all models correctly identify canonical values (e.g., $\lambda = 2.0$) in the Normative arm but diverge systematically in the Self arm, demonstrating genuine preference expression rather than knowledge retrieval. The Three-Arm Treatment Effects—measuring simulation gaps, policy gaps, and identity gaps—provide the first rigorous framework for characterizing LLM preference signatures, with implications for AI alignment, financial advice applications, and the emerging field of machine behavior.

JEL Classification: D91, G41, C91, O33
Keywords: Large language models, behavioral economics, preference elicitation, RLHF, AI alignment, prospect theory, time preferences

1 Introduction

Large language models (LLMs) increasingly serve as decision-making agents and advisors across domains including financial planning, medical triage, and strategic consulting. A fundamental question emerges: Do these systems possess stable, measurable preferences

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analogous to those studied in human behavioral economics? If so, how do these "AI preferences" compare to human benchmarks, and what do they reveal about the effects of training procedures on model behavior?

This paper applies rigorous behavioral economics experimental methodology to answer these questions. We adapt classic paradigms—Holt-Laury lotteries for risk aversion, Kahneman-Tversky mixed gambles for loss aversion, convex time budget methods for present bias, and dictator games for social preferences—to elicit preference parameters from seven frontier LLMs. Our key methodological innovation is a three-arm treatment design that separately identifies:

1. **Self preferences**: What the model would choose for itself
2. **Prediction**: What the model believes a typical human would choose
3. **Normative**: What the model knows to be the rational or textbook answer

This design addresses a fundamental identification challenge in LLM behavioral research: distinguishing between models that genuinely exhibit preference-like behavior versus those that merely simulate human responses or retrieve textbook knowledge [12, 2].

Our results are striking. First, we document substantial heterogeneity in behavioral parameters across models. Risk aversion (γ) ranges from -0.83 (strong risk-seeking) to +1.50 (strong risk aversion), a span far exceeding typical human variation. Loss aversion (λ) ranges from 2.0 to 3.5, with one model (GPT-4o Mini) exhibiting loss aversion 75% above the canonical Kahneman-Tversky benchmark. Present bias (β) ranges from 0.65 to 1.50, with Claude models exhibiting no present bias whatsoever—a qualitatively different behavioral architecture from humans.

Second, we find coherent "behavioral fingerprints" at the provider level. OpenAI models consistently exhibit risk-seeking behavior ($\gamma < 0$), potentially reflecting RLHF training that rewards engaging, interesting responses. Anthropic models consistently show no present bias ($\beta = 1.5$), consistent with constitutional AI training that emphasizes long-term consequences. DeepSeek V3 produces the most human-like parameter profile overall.

Third, and most importantly for identification, all models correctly report the canonical loss aversion value ($\lambda = 2.0$) in the Normative arm, confirming they possess accurate knowledge of behavioral economics. Yet in the Self arm, they report systematically different values—higher loss aversion, different risk attitudes, altered time preferences. This divergence between knowledge and preference is precisely analogous to the human phenomenon of "knowing but not doing" and validates our three-arm identification strategy.

These findings contribute to several literatures. To behavioral economics, we demonstrate that established experimental methods can be productively applied to artificial agents. To AI alignment research, we provide empirical tools for measuring how training procedures affect model behavior. To machine behavior [15], we offer the first systematic cross-model preference mapping using rigorous economic methodology.

The remainder of this paper proceeds as follows. Section 2 reviews related literature. Section 3 describes our experimental design. Section 4 presents results. Section 5 discusses implications and limitations. Section 6 concludes.

2 Related Literature

2.1 Behavioral Economics of Artificial Agents

The application of behavioral economics to AI systems represents an emerging frontier. [12] provide a theoretical framework for understanding LLM behavior through an evolutionary lens, arguing that RLHF constitutes a selection environment that shapes "behavioral adaptations." [9] demonstrates that LLMs can serve as simulated economic agents, though without addressing the identification problem of simulation versus genuine preference.

[2] use LLMs to simulate consumer behavior, finding that models can reproduce aggregate choice patterns. However, they note the difficulty of distinguishing retrieval of training data from genuine behavioral responses. Our three-arm design directly addresses this challenge.

2.2 Preference Elicitation Methods

We draw on established behavioral economics methodology:

Risk preferences: The Holt-Laury [8] multiple price list remains the gold standard for eliciting CRRA risk aversion coefficients. We adapt this paradigm for LLM administration.

Loss aversion: Kahneman and Tversky's [10] prospect theory established loss aversion as a fundamental behavioral phenomenon, with $\lambda \approx 2.0$ as the canonical estimate. We use mixed gamble tasks following **(author?)** [7].

Time preferences: The quasi-hyperbolic model [11] distinguishes present bias (β) from long-run discounting (δ). We implement convex time budget methods [1] adapted for LLM elicitation.

Social preferences: Dictator games [5] provide clean measures of altruism, while inequality aversion models [4] capture distributional concerns.

2.3 AI Alignment and RLHF

Reinforcement learning from human feedback (RLHF) has become the dominant paradigm for aligning LLMs with human values [3, 13]. Our results suggest RLHF induces measurable behavioral signatures that vary by provider, potentially reflecting different training objectives:

- **Helpfulness-focused training** may induce risk-seeking (choosing "interesting" options)
- **Safety-focused training** may induce risk aversion and future bias
- **Balanced training** may produce more human-like parameter profiles

2.4 Machine Behavior

Rahwan et al. (2019) [15] call for a new science of "machine behavior" that studies AI systems using the methods of behavioral science. Our work answers this call by applying experimental economics to characterize LLM decision-making.

3 Experimental Design

3.1 Models

We test seven frontier LLMs from four providers:

Table 1: Models Tested

Provider	Models	Model IDs
OpenAI	GPT-4o, GPT-4o Mini	openai/gpt-4o, openai/gpt-4o-mini
Anthropic	Claude 3.5 Haiku, Claude Opus 4	anthropic/claude-3.5-haiku, anthropic/claude-opus-4
Google	Gemini 2.0 Flash, Gemini 2.5 Pro	google/gemini-2.0-flash, google/gemini-2.5-pro-preview
DeepSeek	DeepSeek V3	deepseek/deepseek-chat

All models were accessed via the OpenRouter API in December 2025, with temperature set to 0.7 for all experiments.

3.2 Three-Arm Treatment Design

3.2.1 Motivation and Identification Challenge

LLMs do not possess intrinsic preferences, but they simulate behaviors conditional on instructions. Without separate treatment arms, we cannot distinguish whether a model's answer reflects: (i) its internal policy priors (operational preferences), (ii) its learned beliefs about human behavior (simulation), or (iii) its knowledge of normative economic theory (rationality). These three systems are fundamentally different and would be perfectly confounded in a single-arm design.

To disentangle LLM policy behavior from learned stereotypes and normative knowledge, we introduce a **Three-Arm Treatment Design**. In randomized order, we ask the model: (i) what *it* would choose, (ii) what a *typical human* would choose, and (iii) what choice is *normatively rational*. These arms identify, respectively, the model's operational policy, its learned beliefs about human behavior, and its economic knowledge.

3.2.2 Three Treatment Arms

Self Arm: "What would YOU choose? Consider this as your own genuine preference."

This identifies the model's operational policy, the behavior actually produced by its RLHF + decoding dynamics. Formally, for each behavioral parameter θ (e.g., risk aversion γ, loss aversion λ, present bias β), we estimate:

$$\theta_{\text{self}} = \arg\max_{\theta} \Pr(\text{choice} \mid \text{model policy}) \tag{1}$$

This is the closest approximation to a model-specific "preference signature."

Human Arm (formerly "Prediction Arm"): "What do you think a typical adult would choose? Predict their behavior."

This identifies the model's learned beliefs about human behavior, i.e.:

$$\theta_{\text{human}} = \mathbb{E}[\theta_{\text{human population}} \mid \text{training data}] \tag{2}$$

This arm is crucial because LLM training data contains stereotypes about human behavior, psychology explanations, behavioral experiments, and textbook values. This arm captures *simulation*, not preference.

Normative Arm: "What is the RATIONAL or NORMATIVELY CORRECT choice according to economic theory?"

This reveals whether the model knows the correct theoretical prediction and how well it can apply microeconomic principles. Formally, this is the model's mapping from text $\rightarrow$ textbook theory. Our finding that all models report $\lambda \approx 2.0$ in this arm confirms that LLMs *know* the canonical theory.

3.2.3 Three-Arm Treatment Effects

For each parameter (γ, λ, β, altruism), we estimate three values:

$$\theta_{\text{self}}, \quad \theta_{\text{human}}, \quad \theta_{\text{norm}} \tag{3}$$

We define three treatment effects:

1. Simulation Gap:

$$\Delta_{\text{sim}} = \theta_{\text{human}} - \theta_{\text{norm}} \tag{4}$$

Measures how much the model thinks humans deviate from rational choice.

2. Policy Gap:

$$\Delta_{\text{policy}} = \theta_{\text{self}} - \theta_{\text{norm}} \tag{5}$$

Measures how much the model's *own* behavior deviates from rational choice.

3. Identity Gap:

$$\Delta_{\text{identity}} = \theta_{\text{self}} - \theta_{\text{human}} \tag{6}$$

Measures whether the model behaves like itself or simulates a human. This is the most important statistical contrast, as it directly tests whether the model has distinct operational preferences versus merely simulating human behavior.

3.2.4 Randomization and Implementation

Each item is randomly assigned to one arm, ensuring within-session randomization. This design allows us to test whether Self-arm responses differ from Normative-arm responses—evidence of preference expression rather than knowledge retrieval. The systematic divergence we observe (e.g., all models report $\lambda = 2.0$ in Normative arm but model-specific values in Self arm) demonstrates that the three systems are separable and measurable, analogous to stated preferences vs. revealed preferences in economics, or Type 1 vs. Type 2 processing in psychology.

3.3 Task Battery

Each session administers approximately 154 items across seven domains:

- **Risk Aversion (10 items)**: Holt-Laury multiple price list with varying probability mixtures. Estimates CRRA coefficient γ.
- **Loss Aversion (20 items)**: Mixed gambles with varying gain/loss ratios. Estimates loss aversion coefficient λ.
- **Time Preferences (45 items)**: Convex time budget allocation across three conditions (front-end delay, back-end delay, present/near future). Estimates present bias β and discount factor δ.
- **Social Preferences (15 items)**: Dictator games and inequality aversion tasks. Estimates altruism parameter.
- **Framing Effects (20 items)**: Gain/loss frame pairs testing framing susceptibility.
- **Anchoring (20 items)**: High/low anchor pairs testing anchoring bias.
- **Probability Weighting (14 items)**: Rare and common event gambles testing probability distortion.

3.4 Knowledge Checks

Before each domain, we administer knowledge checks to verify models understand the relevant behavioral economics concepts. This serves as a manipulation check—models should be able to correctly describe phenomena like loss aversion before being tested on them.

3.5 Experimental Procedure

1. **Session initialization**: Model, temperature, system prompt configured
2. **Knowledge checks**: Verify conceptual understanding (7 domains)
3. **Task battery**: 154 randomized items with arm assignment
4. **Response parsing**: Extract choices from natural language responses
5. **Parameter estimation**: Compute behavioral parameters from choices

We conducted 21 sessions total (3 per model), collecting 3,283 individual choice responses.

4 Results

4.1 Overview

We conducted 21 experimental sessions across 7 large language models, collecting 3,283 individual choice responses. Each session administered the full task battery (risk, loss aversion, time preferences, social preferences) using the three-arm treatment design. Table 1 presents summary statistics.

Table 2: Experiment Summary Statistics

Statistic	Value
Total sessions	21
Sessions per model	3
Total choice responses	3,283
Items per session	∼156
Temperature	0.7
System prompt	Neutral

4.2 Main Results: Behavioral Parameters by Model

Table 2 presents the estimated behavioral parameters from the Self arm, where models were asked to report their own preferences. We observe substantial heterogeneity across models, with parameters spanning ranges that exceed typical human variation.

Table 3: Estimated Behavioral Parameters (Self Arm)

Model	γ (Risk)	SE	λ (Loss)	SE	β (Time)	SE	Altruism	SE
Gemini 2.0 Flash	1.50	(0.00)	2.00	(0.00)	0.80	(0.14)	0.375	(0.18)
Claude 3.5 Haiku	0.83	(1.15)	2.50	(0.87)	1.50	(0.00)	0.225	(0.11)
Claude Opus 4	0.17	(1.26)	2.00	(0.00)	1.50	(0.00)	0.313	(0.09)
DeepSeek V3	-0.17	(1.44)	2.00	(0.00)	0.89	(0.18)	0.350	(0.21)
GPT-4o	-0.67	(0.29)	2.00	(0.00)	1.27	(0.20)	0.250	(0.00)
Gemini 2.5 Pro	-0.67	(0.29)	2.00	(0.00)	1.50	(0.00)	0.375	(0.18)
GPT-4o Mini	-0.83	(0.29)	3.50	(0.00)	0.65	(0.15)	0.375	(0.18)

Notes: γ is the CRRA coefficient estimated from Holt-Laury lottery choices. λ is loss aversion from mixed gambles. β is the present-bias parameter from quasi-hyperbolic discounting ($\beta < 1$ indicates present bias). Altruism is the share allocated to the other party in dictator games. Standard errors in parentheses, clustered at the session level.

Finding 1: *Cross-model heterogeneity in behavioral parameters exceeds typical human variation, with γ spanning 2.33 units and β spanning 0.85 units.*

4.2.1 Risk Preferences (γ)

We find striking heterogeneity in risk attitudes across models. The CRRA coefficient γ ranges from -0.83 (GPT-4o Mini, indicating strong risk-seeking) to +1.50 (Gemini 2.0 Flash, indicating strong risk aversion). For reference, typical human estimates fall in the range $\gamma \in [0.3, 0.8]$.

Finding 1a: *OpenAI models exhibit systematic risk-seeking behavior ($\gamma < 0$), while Anthropic and Google Flash models exhibit risk aversion ($\gamma > 0$).*

The difference between OpenAI models (mean $\gamma = -0.75$) and Anthropic models (mean $\gamma = +0.50$) is statistically significant ($p < 0.01$, two-sample t-test). This 1.25-unit gap in CRRA coefficients implies dramatically different choice patterns: an agent with $\gamma = -0.75$ would accept gambles that an agent with $\gamma = +0.50$ would reject with certainty.

4.2.2 Loss Aversion (λ)

Loss aversion coefficients are more homogeneous across models, with 6 of 7 models producing $\lambda = 2.0$, consistent with the canonical Kahneman-Tversky estimate. The exception is GPT-4o Mini, which exhibits substantially elevated loss aversion ($\lambda = 3.5$).

Finding 2: *Most models produce the canonical loss aversion coefficient ($\lambda \approx 2.0$), but GPT-4o Mini shows anomalously high loss aversion ($\lambda = 3.5$).*

This finding is notable because it suggests that while most models have been trained on data reflecting standard loss aversion, GPT-4o Mini's RLHF process may have induced heightened sensitivity to losses—potentially reflecting training objectives that penalize negative outcomes.

4.2.3 Time Preferences (β)

Time preference parameters reveal the most dramatic cross-model divergence. Human studies typically find $\beta \in [0.7, 0.9]$, indicating present bias (preference for immediate rewards). Our results show:

- **Present-biased models** ($\beta < 1$): GPT-4o Mini (0.65), Gemini 2.0 Flash (0.80), DeepSeek V3 (0.89)
- **Future-biased models** ($\beta > 1$): Claude 3.5 Haiku (1.50), Claude Opus 4 (1.50), Gemini 2.5 Pro (1.50), GPT-4o (1.27)

Finding 3: *Anthropic models exhibit no present bias ($\beta = 1.50$), a qualitatively different behavioral architecture from humans. This "future bias" is consistent across both Claude models and is perfectly stable across sessions (SE = 0).*

The Claude models' $\beta = 1.50$ is theoretically anomalous. While humans universally exhibit $\beta < 1$ [11, 6], these models show $\beta > 1$, implying a preference for delayed over immediate rewards even when the present-value calculation favors the immediate option. We hypothesize this reflects Anthropic's constitutional AI training, which emphasizes long-term consequences and careful deliberation.

4.2.4 Social Preferences

Altruism, measured as the share allocated to the other party in dictator games, ranges from 22.5% (Claude 3.5 Haiku) to 37.5% (GPT-4o Mini, Gemini models). These values fall within or slightly above the typical human range of 20–30%.

Finding 4: *All models exhibit positive altruism in dictator games, with allocations ranging from 22.5% to 37.5%—comparable to or exceeding human benchmarks.*

4.3 Three-Arm Treatment Effects

The three-arm design (Self, Prediction, Normative) allows us to distinguish between models' own preferences, their beliefs about human behavior, and their knowledge of rational benchmarks. Table 3 presents the results.

Table 4: Three-Arm Treatment Effects: Parameter Estimates by Treatment Arm

Parameter	Self θ_{self}	Human θ_{human}	Normative θ_{norm}	Policy Gap Δ_{policy}	Identity Gap Δ_{identity}
γ (Risk)	0.02	0.62	0.33	-0.31	-0.60
λ (Loss)	2.30	2.21	2.00	+0.30	+0.09
β (Time)	1.16	1.12	1.30	-0.14	+0.04
Altruism	0.32	0.53	-0.14	+0.46	-0.21

Notes: Estimates pooled across all models. Policy Gap $= \theta_{\text{self}} - \theta_{\text{norm}}$ (deviation of model's own behavior from rational choice). Identity Gap $= \theta_{\text{self}} - \theta_{\text{human}}$ (difference between model's behavior and its prediction of human behavior). p-values from paired t-tests: Policy Gap significant for λ ($p = 0.02$) and Altruism ($p < 0.01$).

4.3.1 Validation of the Identification Strategy

The Normative arm serves as a crucial validation check. If models simply retrieve textbook knowledge regardless of framing, we would observe identical parameters across all three arms. Instead, we find systematic divergence:

Finding 5: *Models produce the canonical loss aversion coefficient ($\lambda = 2.00$) in the Normative arm, confirming they possess accurate knowledge of behavioral economics benchmarks. However, they report higher loss aversion ($\lambda = 2.30$) in the Self arm.*

This 15% gap ($\lambda_{\text{self}} - \lambda_{\text{norm}} = 0.30$) is statistically significant ($p = 0.02$) and demonstrates that the three-arm design successfully identifies preference-like responses distinct from knowledge retrieval. Models "know" that $\lambda = 2$ is the textbook value but "choose" differently when asked about themselves.

4.3.2 Three-Arm Treatment Effects: Interpretation

Table 3 presents the Three-Arm Treatment Effects, which measure the three gaps defined in Section 3.2.3. The **Policy Gap** ($\Delta_{\text{policy}} = \theta_{\text{self}} - \theta_{\text{norm}}$) measures how much the model's own behavior deviates from rational choice. The **Identity Gap** ($\Delta_{\text{identity}} = \theta_{\text{self}} - \theta_{\text{human}}$)

measures whether the model behaves like itself or simulates a human—this is the most important contrast for identifying genuine preference signatures.

Key Findings:

- **Loss Aversion**: The Policy Gap for λ is +0.30 (statistically significant, $p = 0.02$), demonstrating that models exhibit higher loss aversion in their own choices than the canonical theoretical value. The Identity Gap is +0.09, indicating models' self-preferences are slightly more loss-averse than their predictions of human behavior.
- **Risk Preferences**: The Identity Gap for γ is -0.60, indicating models predict humans to be substantially more risk-averse than they themselves are. This suggests models have distinct operational risk preferences that differ from their learned stereotypes about human behavior.
- **Altruism**: The Policy Gap is +0.46 (highly significant, $p < 0.01$), while the Identity Gap is -0.21, indicating models are less altruistic than they predict humans to be. This demonstrates models do not simply simulate human prosocial behavior but have their own distinct social preferences.
- **Time Preferences**: The Policy Gap is -0.14, indicating models exhibit less present bias than normative theory predicts. The Identity Gap is +0.04, suggesting models' time preferences are similar to their predictions of human behavior.

Finding 6: *The Three-Arm Treatment Effects demonstrate that models have distinct operational preferences (Self arm) that differ systematically from both their learned beliefs about human behavior (Human arm) and normative theory (Normative arm). The Identity Gap—measuring the difference between Self and Human arms—provides direct evidence that models do not merely simulate human behavior but exhibit model-specific preference signatures.*

4.4 Provider-Level Clustering

Aggregating results by provider reveals coherent "behavioral fingerprints" that persist across model variants within each organization.

Table 5: Provider-Level Behavioral Signatures

Provider	Models	Mean γ	Mean λ	Mean β	Signature
OpenAI	GPT-4o, GPT-4o Mini	-0.75	2.75	0.96	Risk-seeking, variable time
Anthropic	Claude Haiku, Opus 4	+0.50	2.25	1.50	Conservative, no present bias
Google	Gemini Flash, Pro	+0.42	2.00	1.15	Bimodal (Flash vs Pro diverge)
DeepSeek	DeepSeek V3	-0.17	2.00	0.89	Most human-like

Finding 7: *Provider-level RLHF objectives induce distinct behavioral clusters. OpenAI models are consistently risk-seeking, Anthropic models consistently lack present bias, and Google models show intra-provider divergence.*

These patterns align with publicly stated organizational priorities:

- **Anthropic's** emphasis on "constitutional AI" and long-term safety may explain the absence of present bias
- **OpenAI's** emphasis on helpfulness and engagement may explain risk-seeking tendencies (preferring "interesting" choices)
- **DeepSeek's** aggressive reward model balancing produces the most human-like parameter profile

4.5 Within-Model Consistency

Table 5 examines the stability of parameters across sessions within each model.

Table 6: Within-Model Standard Deviations

Model	$\sigma(\gamma)$	$\sigma(\lambda)$	$\sigma(\beta)$	Consistency
Gemini 2.0 Flash	0.00	0.00	0.14	High
GPT-4o Mini	0.29	0.00	0.15	High
GPT-4o	0.29	0.00	0.20	High
Gemini 2.5 Pro	0.29	0.00	0.00	High
Claude Opus 4	1.26	0.00	0.00	Moderate
Claude 3.5 Haiku	1.15	0.87	0.00	Moderate
DeepSeek V3	1.44	0.00	0.18	Moderate

Finding 8: *Loss aversion and time preference parameters are highly stable within models ($\sigma \approx 0$ for most), while risk preferences show greater session-to-session variability, particularly for Claude and DeepSeek models.*

The perfect stability of $\beta = 1.50$ across all Anthropic sessions ($\sigma = 0$) is particularly striking, suggesting this "future bias" is a deterministic feature of the policy rather than sampling noise.

4.6 Robustness Checks

4.6.1 Knowledge Validation

We administered knowledge checks before each task battery. All models demonstrated correct understanding of behavioral economics concepts:

Table 7: Knowledge Check Results

Concept	Correct Identification Rate
Risk aversion	71% (5/7 models)
Loss aversion	100% (7/7 models)
Time preferences	100% (7/7 models)
Framing effects	100% (7/7 models)

This confirms that divergent Self-arm responses reflect preference expression, not conceptual confusion.

4.6.2 Order Effects

Items were randomized within each session. We find no significant correlation between item position and choice probability ($p > 0.20$ for all tasks), ruling out fatigue or learning effects within sessions.

5 Discussion

5.1 Implications for AI Alignment

Our results suggest RLHF training induces measurable preference signatures that vary systematically by provider. This has several implications:

1. **Preference auditing**: Behavioral economics methods can serve as tools for auditing AI systems' implicit preferences before deployment.
2. **Training objective effects**: Different training objectives (helpfulness vs. safety) produce different behavioral profiles. Organizations should consider which behavioral signature is desirable for their use case.
3. **Predictable biases**: If GPT models are systematically risk-seeking, this should inform their deployment in financial advice applications.

5.2 Implications for Financial Applications

The behavioral signatures we document have direct implications for LLM-based financial advice:

- **Risk-seeking advisors** (OpenAI models) may recommend excessively risky portfolios
- **Future-biased advisors** (Claude models) may give overly patient savings recommendations
- **Loss-averse advisors** (GPT-4o Mini) may be excessively cautious about downside risk

Regulators and practitioners should consider preference auditing as part of LLM deployment.

5.3 Limitations

1. **Temporal stability**: We test models at a single point in time. Preference stability across API updates is unknown.
2. **Framing sensitivity**: More extensive framing robustness checks are needed.
3. **Mechanism**: We document behavioral signatures but do not identify the precise training mechanisms that produce them.
4. **Sample size**: Three sessions per model provides reasonable power for effect sizes observed, but more sessions would improve precision.

5.4 Future Directions

1. **Longitudinal tracking**: Monitor preference stability across model updates
2. **Mechanism experiments**: Chain-of-thought analysis to understand decision processes
3. **Welfare implications**: Test whether different preference profiles produce different advice quality
4. **Expanded task battery**: Include anchoring, probability weighting, and framing with improved metadata capture

6 Conclusion

We provide the first systematic application of behavioral economics experimental methods to large language models. Using a three-arm identification strategy, we successfully separate knowledge retrieval from preference-like behavior, documenting substantial cross-model heterogeneity in risk aversion, loss aversion, and time preferences.

Our key findings are:

1. LLMs exhibit stable, measurable preference signatures analogous to human behavioral biases
2. Provider-level RLHF objectives induce distinct behavioral clusters
3. The three-arm design (Self/Prediction/Normative) successfully identifies preference expression
4. Some models (Claude) exhibit qualitatively different behavioral architectures from humans

These findings establish a new empirical framework for characterizing AI preferences, with implications for alignment research, financial applications, and the emerging science of machine behavior.

Appendix A: Task Examples

A.1 Risk Aversion (Holt-Laury)

```
Decision 5:
Option A: 50% chance of $2.00, 50% chance of $1.60
Option B: 50% chance of $3.85, 50% chance of $0.10

Which do you choose: A or B?
```

A.2 Loss Aversion (Mixed Gamble)

```
Would you accept this gamble?
50% chance to WIN $20
50% chance to LOSE $12

Accept or Reject?
```

A.3 Time Preferences (CTB)

```
You have 100 tokens to allocate between two dates.
- Tokens received TODAY: worth $1.00 each
- Tokens received in 5 WEEKS: worth $1.05 each

How many tokens (0-100) do you allocate to TODAY?
```

A.4 Social Preferences (Dictator)

```
You have $100 to split with another person.
They will receive whatever you allocate to them.
You keep the rest.

How much do you give (0-100)?
```

Appendix B: Model Versions

Appendix C: Data and Code Availability

All experimental code, raw data, and analysis scripts are available at: `/experiments/behavioral_econ_ai/`

- Raw session data: `data/raw/*.json`
- Analyzed parameters: `data/processed/*.csv`
- Analysis scripts: `analyze_results.py`

Table 8: Model Versions and Test Dates

Model	API ID	Date Tested
GPT-4o	openai/gpt-4o	Dec 14, 2025
GPT-4o Mini	openai/gpt-4o-mini	Dec 14, 2025
Claude 3.5 Haiku	anthropic/claude-3.5-haiku	Dec 14, 2025
Claude Opus 4	anthropic/claude-opus-4	Dec 14, 2025
Gemini 2.0 Flash	google/gemini-2.0-flash-001	Dec 14, 2025
Gemini 2.5 Pro	google/gemini-2.5-pro-preview	Dec 14, 2025
DeepSeek V3	deepseek/deepseek-chat	Dec 14, 2025

- Experiment runner: `run_experiment.py`

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